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Microscopic cluster study of the ^{7}Be(p,γ)^{8}B and ^{17}F(p,γ)^{18}Ne reactions at astrophysical energies

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The ^{7}Be(p,γ)^{8}B and ^{17}F(p,γ)^{18}Ne reactions are analyzed in a microscopic cluster model. The ^{7}Be wave functions are defined by an α+^{3}He cluster structure, including distortion effects. We find slight differences with respect to a previous three-cluster calculation. For ^{17}F(p,γ)^{18}Ne, the non-resonant ($E1$) as well as the resonant contributions ($M1, E2$) of the astrophysical S-factor are determined. We pay a special attention to the properties of the 3_1^+ resonance, crucial for astrophysics.

1. INTRODUCTION

The ^{7}Be(p,γ)^{8}B plays a crucial role in the solar-neutrino problem [1]. Many direct as well as indirect measurements have been performed in order to reduce the uncertainties on the S-factor at zero energy (see ref. [2] for an overview). As a high precision is required for $S(0)$, the extrapolation down to astrophysical energies should be done very carefully. Current experiments are performed in a limited energy range, which requires the use of a theoretical model to derive $S(0)$. The reliability of the model can be tested in the energy range where data exist, which provides some "confidence level" on the extrapolation. In most experiments, a microscopic cluster model [3] (hereafter referred to as DB94) is used for the extrapolation. This model takes account of the ^{7}Be deformation, of inelastic and rearrangement channels, and has been tested with spectroscopic properties of ^{8}B and ^{8}Li, as well as with the ^{7}Li(n,γ)^{8}Li mirror cross section.

The knowledge of the ^{17}F(p,γ)^{18}Ne reaction rate is important for the understanding of the energy production and of the nucleosynthesis in novae [4]. The energy range characteristic of novae and x-ray bursts is 0.3 MeV for a typical temperature $T = 0.5$ GK. Until now, a direct measurement of the cross section down to these energies has not been performed. In the literature, the reaction rate is computed as a sum of a direct-capture term and of a resonant contribution, including the 1_1^-, 3_1^+ and 0_3^+ resonances located just above the ^{17}F + p threshold (see references in ref. [5]). It is now well established that the 3_1^+ level dominates the rate at high temperatures since it is the only $s-$wave resonance at low energies. A first approximation of the energy has been obtained from the mirror level

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in ^{18}O. More recently, the 3_1^+ energy and the associated proton width have been measured by Bardayan *et al.* [5]. However, the value of the gamma width remains experimentally unknown and is estimated by shell-model calculations. At low temperatures, the non-resonant contribution is dominant but is evaluated with simple models [6] which need confirmation from more elaborate theories.

Both reactions are investigated here within the Generator Coordinate Method (GCM). This microscopic method provides a "basic" description of a A-nucleon system, as the whole information is obtained from a nucleon-nucleon interaction. Many applications have been considered in recent years (see references in ref. [7]).

2. OUTLINE OF THE MICROSCOPIC MODEL

The A-body Hamiltonian is given by

$$H = \sum_{i=1}^{A} T_i + \sum_{i<j}^{A} V_{ij}, \tag{1}$$

where T_i is the kinetic energy of nucleon i, and V_{ij} a nucleon-nucleon interaction. The total wave functions are defined from cluster wave functions ϕ_1 and ϕ_2 of the colliding nuclei. We have, in a schematic notation,

$$\Psi = \mathcal{A}\,\phi_1\,\phi_2\,g(\boldsymbol{\rho}), \tag{2}$$

where $g(\boldsymbol{\rho})$ is the relative function depending on the relative coordinate $\boldsymbol{\rho}$. It is determined from the Schrödinger equation. In (2), $\mathcal{A}$ is the antisymmetrization operator which ensures the Pauli principle to be satisfied. Projection over good quantun numbers is performed exactly. The reliability on the model mostly depends on the accuracy of the internal wave functions ϕ_1 and ϕ_2. Many efforts have been done to go beyond the simple shell-model approximation: multicluster description [3], monopole distortion [8] and extended shell model developments [9] aim at improving the wave functions (2). In the GCM, definition (2) is applied to bound states as well as to scattering states.

3. THE ^{7}Be(p,γ)^{8}B REACTION

The calculation of DB94 presents two shortcomings. First the ^{7}Be nucleus is assumed to be "frozen" during the collision. In other words, a single generator coordinate R_2 was used (see Fig. 1). On the other hand, the nucleon-nucleon interaction is the Volkov force (V2, ref. [10]) only. These approximations are quite reasonable, but the $S(0)$ value of DB94 is 29 eV.b, which is significantly larger than current values. Although the main output of DB94 is the energy dependence, the conditions should be questioned owing to the high accuracy needed for the ^{7}Be(p,γ)^{8}B reaction. This prompted us to improve DB94 by allowing ^{7}Be to be distorted: the ^{7}Be nucleus is described here by four R_2 values (see fig.1). The ^{7}Be$(3/2^-, 1/2^-, 7/2^-, 5/2^-)$+p and ^{5}Li+^{3}He channels are included. In addition, the analysis is complemented by the use of the Minnesota force (MN, ref. [11]), known to be better adapted to low-mass systems. We have computed spectroscopic properties of ^{8}B, which do not significantly differ from DB94.

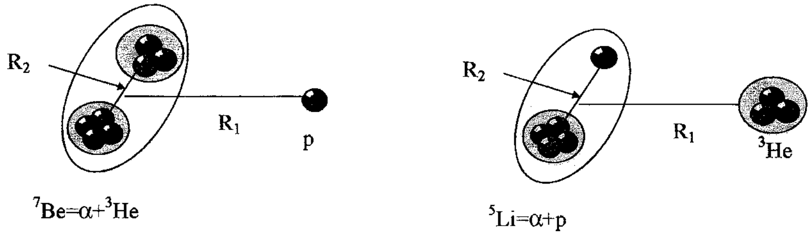

Figure 1. Three-cluster configuration of ^{8}B.

The ^{7}Be(p,γ)^{8}B S-factor is shown in Fig. 2, where we have omitted the data of Kavanagh [12], most likely affected by a normalization problem. We only show the non-resonant E1 component. The role of the ^{7}Be distortion is illustrated by comparing DB94 with the present V2 result: the amplitude is reduced by 10%. With the MN force we have, without any renormalization, $S(0) = 23$ eV.b.

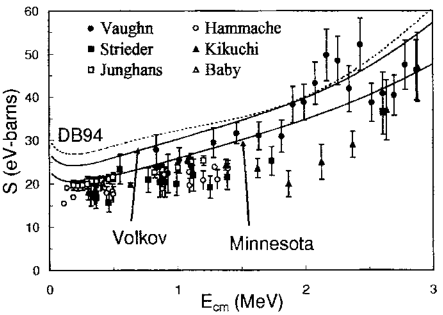

Figure 2. ^{7}Be(p,γ)^{8}B S-factor (E1 term) with data from refs. [13–18]. The results of DB94 are shown as a dashed line.

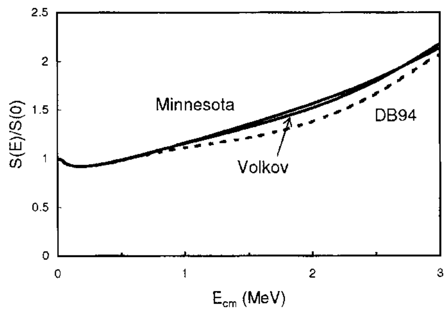

Figure 3. ^{7}Be(p,γ)^{8}B S-factor (E1 contribution) normalized at $E = 0$.

As discussed in DB94, a cluster model provides an upper bound of the capture cross section. The "exact" ^{8}B wave function should contain many other configurations (other arrangements, 4 clusters, 5 clusters, etc.). Accordingly, the capture cross section which, up to the electromagnetic operator, is nothing but the overlap between the initial ^{7}Be+p and final ^{8}B wave functions, is in general overestimated by a cluster model. This overestimation factor decreases as the model is improved.

Even if the present MN calculation provides a significant improvement with respect to DB94, it is likely that the energy dependence is still more accurate than the normalization itself. Figure 3 shows the energy dependence, normalized to zero energy. The new calculations with V2 and MN are almost undistinguishable; below 0.6 MeV, DB94 gives the same results. However the difference with DB94 reaches about 10% near 2 MeV; the slope of the S-factor between 1.5 and 3 MeV is weaker than in DB94. Although this modification is minor, it may affect current extrapolations, especially for breakup experiments, usually performed above 1 MeV.

4. THE ^{17}F(p,γ)^{18}Ne REACTION

The present work aims at studying the ^{17}F(p,γ)^{18}Ne radiative capture reaction with an Extended Two-Cluster Model (ETCM). In order to investigate the ^{17}F(p,γ)^{18}Ne reaction, the ETCM wave functions of ^{18}Ne are defined as a combination of ^{17}F + p and α+^{14}O channel functions. The ^{17}F and the ^{14}O nuclei are described by s, p and sd (for ^{17}F) harmonic oscillator shells. This framework allows the description of the ^{17}F$(5/2^+, 1/2^+, 3/2^+)$+p and ^{14}O$(0^+, 2^+) + \alpha$ channels. Details can be found in ref. [19].

In Table 1, we sum up the properties of the 3_1^+ state, relevant for astrophysics. The proton width (21 keV) is in fair agreement with the experimental value (18 ± 3 keV) [5]. Gamma widths used in the recent evaluations of the reaction rate [5] are taken from the work of Garcia *et al* [6].

Table 1
$M1$ reduced transition probabilities (in W.U.) and Γ_γ widths in (meV) from the 3_1^+ states in ^{18}Ne.

$J_i^{\pi_i}$	$J_f^{\pi_f}$	$B(M1)^{GCM}$	Γ_γ (Ref. [6])	Γ_γ^{GCM}
3_1^+	2_1^+	7.97×10^{-2}	25 ± 16	30
3_1^+	2_2^+	1.99×10^{-1}	3.8 ± 3.1	3.1
3_1^+	4_1^+	6.97×10^{-4}	0.8 ± 0.8	0.02

The astrophysical S-factor is calculated for the $E1$, $E2$, and $M1$ multipolarities. In each case, the four 0_1^+,2_1^+, 2_2^+ and 4_1^+ bound states are considered. The J values for scattering states are limited to 4, which corresponds to $\ell = 2$. The total and partial S-factors are displayed in Fig. 4.

5. CONCLUSION

The ^{7}Be(p,γ)^{8}B S-factor of DB94 has been reanalyzed with improved conditions. The overall normalization is now very close to values deduced from experiment. The energy dependence is slightly different from DB94 in the energy range 1.5 − 3.0 MeV, which could affect extrapolations of data obtained in this energy range.

In ^{17}F(p,γ)^{18}Ne, the gamma width of the 3_1^+ resonance is $\Gamma_\gamma = 33$ meV, a value similar to those used in the literature, but the uncertainties considered up to now (66%) could be significantly reduced. The ^{17}F(p,γ)^{18}Ne reaction rate (see ref. [19]) is found larger by

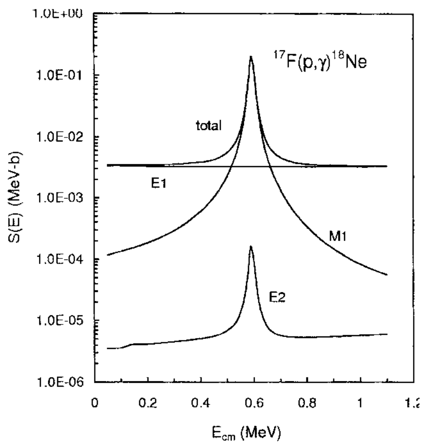

Figure 4. ^{17}F(p,γ)^{18}Ne astrophysical S-factor with the contribution of different multipolarities.

30% than previously assumed [5]. In ref. [19], we also give ^{18}Ne vertex constants, which could be used to estimate the non-resonant part of the cross section.

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